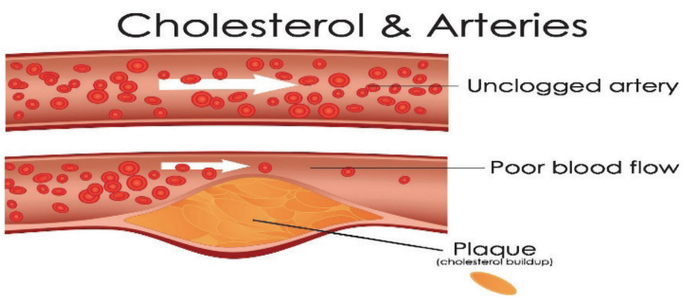
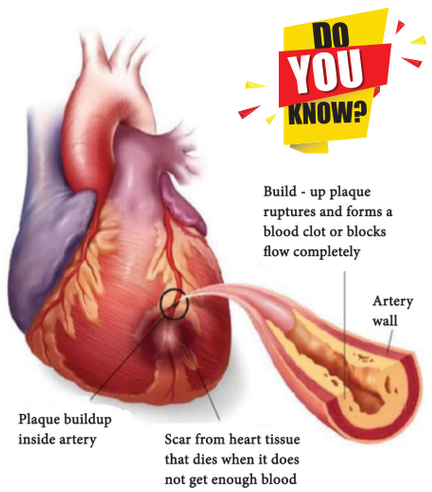
6. Cholesterol when oxidized under suitable conditions, undergoes rapid oxidation to form a ketone called cholestenone.

7. The hydroxyl group of cholesterol readily forms ester with fatty acids like stearic acid.

Atherosclerosis is a hardening of an artery. Meaning *arteria*- 'artery', *sklerosis*–'hardening'. It is asymptomatic for decades until enough narrowing or closure of an artery, due to clots, occurs. And impedes blood flow to different organs enough to induce symptoms.

Typically, atherosclerosis begins at young age, as a thin layer of white-yellowish streaks with the inner layers of the artery walls and progresses from there.





Importance of Cholesterol:

Cholesterol is vital to cell membrane structure, and functions as a precursor to fat-soluble vitamins and steroid hormones

- It is also a key regulator of membrane fluidity in animals.
- It forms lipid rafts when it complexes with certain phospholipids, which makes the membranes less fluid and less subject to phase transitions. This also increases the permeability of the cell membrane to hydrogen and sodium ions.
- It also assists the liver in the manufacture of bile acids, which is essential for digestion and absorption of fat-soluble vitamins such as vitamin A, D, E and K.
- It helps in maintenance of our body temperature and protects our internal organs.
- In pharmaceutical industries, cholesterol is used in the manufacture of steroid hormones and vitamin D.

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